

Projectile motion

Pearson Edexcel AS Level Physics

Name

Class

Date

4

questions

23

total marks

Instructions

Use black ink or ball-point pen.

Answer all questions in the spaces provided; there may be more space than you need.

Show your working in calculations and include units where appropriate.

The marks for each question are shown in brackets.

Questions taken from

Physics · Specimen 2015 · Paper 01

Physics · May/June 2017 · Paper 01

Physics · May/June 2019 · Paper 01

Physics · May/June 2018 · Paper 01

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- 1 The photograph shows a small plastic container, its lid and some pellets of frozen carbon dioxide, known as dry ice. When at room temperature the dry ice gradually changes state directly from solid to gas.



Dry ice is placed in the container and the lid is put on. The container is turned upside down and placed on the floor. After a few minutes the pressure of the gas causes the container to fly into the air, leaving the lid and some dry ice behind.

A student investigated the motion of the container.

- (a) The student obtained measurements of the maximum height reached by the container for a particular initial mass of dry ice. The student determined that the maximum height was 2.5 m.

Calculate the initial speed of the container.

(2)

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Initial speed =

- (b) The student investigated how the maximum horizontal distance travelled by the container varies with launch angle.

Calculate the maximum horizontal distance the container would travel if launched at an initial speed of 6.5 m s^{-1} at an angle of 20° to the vertical.

(5)

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Maximum horizontal distance =

(Total for Question 1 = 7 marks)

Source: Physics · Specimen 2015 · Paper 01 · Q15

- 2 A ball is kicked, giving it an initial velocity v at an angle of 35° to the vertical. The ball remains in the air for a time t .

Which of the following gives the distance travelled over the ground?

☐ A $vt \cos 35^\circ$

☐ B $\frac{v \cos 35^\circ}{t}$

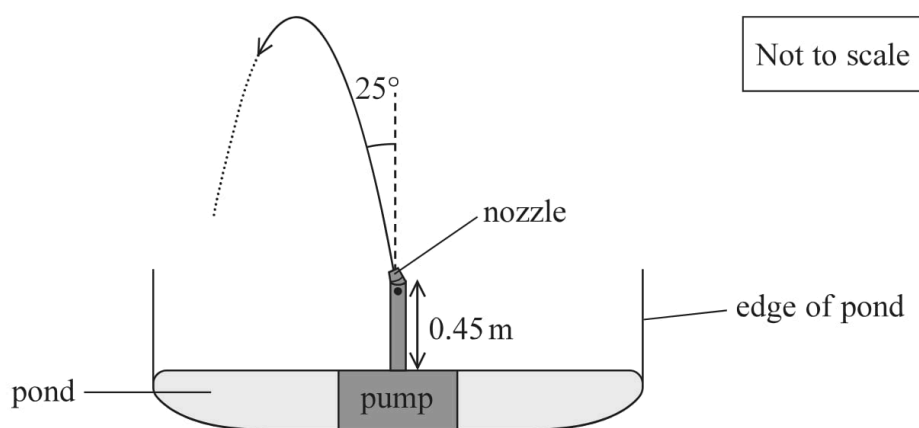
☐ C $vt \sin 35^\circ$

☐ D $\frac{v \sin 35^\circ}{t}$

(Total for Question 2 = 1 mark)

Source: Physics · May/June 2017 · Paper 01 · Q6

- 3 A garden pond contains a water fountain. The fountain consists of a pump and nozzle and is used to create a jet of water which falls back into the pond as shown. The top of the edge of the pond is level with the nozzle.



The water is lifted through a height of 0.45 m before it leaves the nozzle at an angle of 25° to the vertical.

- (a) The efficiency of the pump is 76%.

Show that the water leaves the nozzle at a speed of about 8 m s^{-1} .

mass of water leaving the pump each second = 3.5 kg s^{-1}

power of pump = 160 W

(5)

- (b) Determine the minimum horizontal distance that the nozzle should be placed from the edge of the pond so that the water falls back into the pond. It can be assumed that air resistance is negligible.

(4)

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Minimum horizontal distance =

- (c) The minimum horizontal distance calculated in (b) is an overestimate because air resistance was ignored.

Explain another reason for the distance calculated to be an overestimate.

(2)

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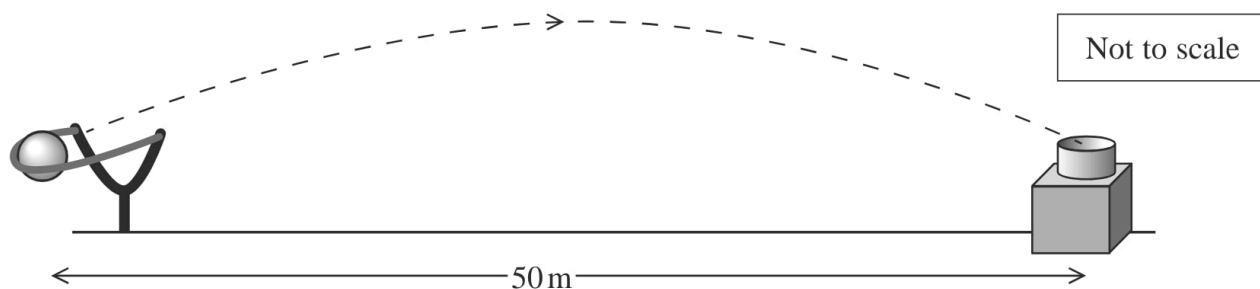
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(Total for Question 3 = 11 marks)

Source: Physics · May/June 2019 · Paper 01 · Q12

- 4 A fairground game requires the player to catapult a ball towards a target to score points.

The ball is required to reach a target a horizontal distance of 50 m away, at the same vertical height, as shown.



- (a) The time taken for the ball to reach the target is 2.0 s.

Calculate the angle to the horizontal at which the ball is launched.

(4)

Angle to the horizontal =

(Total for Question 4 = 4 marks)

Source: Physics · May/June 2018 · Paper 01 · Q11

TOTAL FOR WORKSHEET = 23 MARKS

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Answers

1 Physics · Specimen 2015 · Paper 01 · Q15

(a) Use of $v^2 = u^2 + 2as$ (1) OR use of $\frac{1}{2}mv^2 = mgh$ (1) (2)

Initial speed = 7.0 m s^{-1} (1)

Example of calculation: $v = 0$, $a = -9.81 \text{ m s}^{-2}$, $s = 2.5 \text{ m}$; $u^2 = -2as = 2 \times 9.81 \times 2.5 = 49 \text{ m}^2 \text{ s}^{-2}$; $u = 7.0 \text{ m s}^{-1}$

Alternative: $v = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 2.5} = 7.0 \text{ m s}^{-1}$

(b) Use of trig function to find v vertical (1) (5)

Use of trig function to find v horizontal (1)

Use of equation of motion to find time of flight (1)

Use of equation of motion to find distance (1)

Horizontal distance = 2.7 m (1)

Example of calculation (as printed in the mark scheme): vertical velocity = $6.5 \text{ m s}^{-1} \sin 20 = 2.22 \text{ m s}^{-1}$; time of flight using $v = u + at$: $-2.22 = 2.22 + (-9.81 \times t)$, $t = 0.45 \text{ s}$;
horizontal velocity = $6.5 \text{ m s}^{-1} \cos 20 = 6.11 \text{ m s}^{-1}$; horizontal distance $s = ut = 6.11 \text{ m s}^{-1} \times 0.45 \text{ s} = 2.7 \text{ m}$

2 Physics · May/June 2017 · Paper 01 · Q6

C (1)

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- (a) (5)
- Use of efficiency = $\frac{\text{useful power output}}{\text{total power input}}$ (1)
 - Use of $\Delta E_{\text{grav}} = mg\Delta h$ (1)
 - Use of $E_k = \frac{1}{2}mv^2$ (1)
 - Use of (output) power = $\frac{\Delta E_{\text{grav}}}{1\text{ s}} + \frac{E_k}{1\text{ s}}$ (1)
 - $v = 7.8\text{ m s}^{-1}$ (1)

Example: Output power = $0.76 \times 160\text{ W} = 121.6\text{ W}$; $121.6 = (3.5)(9.81)(0.45) + \frac{1}{2}(3.5)v^2$;
 $v = 7.78\text{ m s}^{-1}$

- (b) (4)
- Use of trig to determine the vertical or horizontal component of the initial velocity (1)
 - Use of equation(s) of motion to determine time the water is in air (1)
 - Use of $s = vt$ (1)
 - $s = 4.8\text{ m}$ (1)

Allow ecf from (a). (Using show that value, $s = 5.0\text{ m}$)

Example: $u_h = 7.8 \sin 25 = 3.30\text{ m s}^{-1}$; $u_v = 7.8 \cos 25 = 7.07\text{ m s}^{-1}$; $0 = 7.07t + \frac{1}{2}(-9.81)t^2$, $t = 1.44\text{ s}$; $s = 1.44 \times 3.30 = 4.75\text{ m}$

- (c) (2)
- There is friction between the water and the pipes (1)
 - This will reduce the (initial) velocity of the water (1)

MP2 is dependent on MP1.

Use of $v = s/t$ for the horizontal motion, or see $u_h = 25\text{ m s}^{-1}$ (1) (4)

Use of $s = ut + \frac{1}{2}at^2$ with $s = 0$, or $v = u + at$ with $t = 1.0\text{ s}$, or see $u_v = 9.81\text{ m s}^{-1}$ (1)

Combining of horizontal velocity and vertical velocity expressions, or see $\tan \theta = \frac{9.81}{25}$ (1)

$\theta = 21^\circ$ (1)

Example of calculation: $u_h = \frac{50\text{ m}}{2.0\text{ s}} = 25\text{ m s}^{-1}$; $0 = u_v(2.0\text{ s}) + \frac{1}{2}(-9.81\text{ m s}^{-2})(2.0\text{ s})^2$, $u_v = 9.81\text{ m s}^{-1}$; $\tan \theta = \frac{9.81}{25}$, $\theta = 21.4^\circ$